Cuantificación de Incertidumbre en Clasificación Histopatológica

Elena Ardura Carnicero
Universidad Pontificia Comillas
202111181@alu.comillas.edu

Victoria García Martínez-Echevarría
Universidad Pontificia Comillas
202112284@alu.comillas.edu

Abstract

La adopción clínica de sistemas de aprendizaje profundo en anatomía patológica se ve limitada no tanto por la precisión de los modelos, sino por la incapacidad de estos de señalar cuándo no están seguros de sus predicciones. En este trabajo se comparan empíricamente tres modelos (dos de ellos con cuantificación de incertidumbre) que parten del mismo clasificador de referencia (ResNet18 preentrenada) aplicado al dataset *Histopathologic Cancer Detection*: (i) una red determinista entrenada como estimador máximo a posteriori (MAP); (ii) una aproximación de Laplace *post-hoc* sobre la última capa, con Hessiano factorizado por Kronecker (KFAC) y precisión del prior optimizada por verosimilitud marginal; y (iii) Monte Carlo Dropout con $T{=}50$ pasadas estocásticas en inferencia. Los tres modelos se evalúan en términos de discriminación (AUC-ROC, F1), calibración probabilística (Expected Calibration Error, Brier Score, log-verosimilitud negativa) y utilidad clínica mediante un análisis de triaje basado en incertidumbre epistémica.

1 Introducción

1.1 Motivación clínica y valor de negocio

La detección automática de células tumorales en parches de tejido histopatológico es uno de los casos de uso más consolidados del aprendizaje profundo en medicina. Los modelos convolucionales actuales alcanzan sistemáticamente un AUC superior a $0{,}95$ sobre benchmarks públicos, lo que ha motivado su integración progresiva en flujos de cribado asistido por ordenador. Sin embargo, este rendimiento oculta un problema estructural: un clasificador determinista emite una probabilidad puntual $\hat{p}(y{=}1 \mid \mathbf{x})$ sin ninguna cuantificación de la fiabilidad de esa estimación. Un caso morfológicamente ambiguo y un caso prototípico pueden recibir la misma probabilidad sin que el sistema distinga entre ellos.

Esta limitación es grave en dominios donde el coste de un falso negativo es desproporcionado y crítico, como puede ser la oncología. En este contexto, el enfoque bayesiano aporta una solución práctica: en lugar de aprender un único conjunto de pesos $\boldsymbol{\omega}^*$, el objetivo es inferir una distribución posterior $p(\boldsymbol{\omega} \mid \mathcal{D})$ sobre la que se marginaliza al predecir. La varianza de dicha distribución predictiva posibilita la toma de decisiones diferenciadas: cuando la incertidumbre supera un umbral preconfigurado τ_u, la muestra se deriva a revisión por un patólogo.

Esta regla operativa es la traducción directa del valor de negocio de estos modelos: no se trata solo de mejorar la precisión global, sino de identificar casos inciertos para que un experto (humano) los revise, reduciendo así el riesgo de falsos negativos sin sobrecargar al patólogo con casos típicos. En un despliegue real, las métricas online asociadas serían la tasa de derivación a revisión humana, la frecuencia con la que el patólogo rechaza la recomendación del modelo teniendo visibilidad de la incertidumbre (*user trust*), y la *reducción de falsos negativos* de la combinación modelo-patólogo frente a un patólogo sin apoyo algorítmico.

Probabilistic Artificial Intelligence – MsC in Advanced Artificial Intelligence, Universidad Pontificia Comillas.

1.2 Objetivos y contribuciones

Este trabajo persigue tres objetivos concretos. Primero, comparar cuantitativamente el rendimiento predictivo y la calibración de dos métodos de inferencia aproximada ampliamente utilizados en la literatura: la aproximación de Laplace (**?**) y Monte Carlo (MC) Dropout (**?**). En ambos casos, se parte de un mismo clasificador fijo (ResNet18 preentrenada) para aislar el efecto del método probabilístico. Segundo, analizar y aplicar la descomposición de la incertidumbre predictiva en sus componentes aleatórica y epistémica para definir una política de triaje. Tercero, estudiar los recursos computacionales de cada enfoque (tiempo de entrenamiento, latencia de inferencia, memoria) para contemplar un posible despliegue. El código, los pesos entrenados y una demostración interactiva en Streamlit están disponibles en este repositorio, reproducible con semilla aleatoria fija.

2 Formulación del problema

2.1 Modelo probabilístico

Disponemos de un conjunto de datos (dataset *Histopathologic Cancer Detection*) $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ con imágenes $\mathbf{x}_i \in \mathbb{R}^{3\times 96\times 96}$ y etiquetas binarias $y_i \in \{0, 1\}$ que indican la presencia o ausencia de tejido tumoral la región central (32x32) de la imagen. Modelizamos la verosimilitud como categórica sobre dos clases,

$$p(y \mid \mathbf{x}, \boldsymbol{\omega}) \;=\; \mathrm{Cat}\big(y \,\big|\, \mathrm{softmax}(f_{\boldsymbol{\omega}}(\mathbf{x}))\big), \tag{1}$$

donde $f_{\boldsymbol{\omega}} : \mathbb{R}^{3\times 96\times 96} \to \mathbb{R}^2$ es una red neuronal con pesos $\boldsymbol{\omega} \in \mathbb{R}^P$. Aunque el problema es binario, tomamos la parametrización *softmax* de dos dimensiones por cuestiones de compatibilidad con la librería `laplace-torch` (que asume salida categórica) y con el mecanismo de descomposición de la incertidumbre explicado en la SECCION4!.

Asumimos un prior gaussiano isotrópico sobre los pesos,

$$p(\boldsymbol{\omega}) \;=\; \mathcal{N}\big(\boldsymbol{\omega} \,\big|\, \mathbf{0}, \lambda^{-1}\mathbf{I}\big), \tag{2}$$

con precisión $\lambda > 0$. Para obtener la distribución predictiva bayesiana para una nueva observación $\mathbf{x}^\star$ necesitamos marginalizar sobre el posterior,

$$p(y^\star \mid \mathbf{x}^\star, \mathcal{D}) \;=\; \int p(y^\star \mid \mathbf{x}^\star, \boldsymbol{\omega})\, p(\boldsymbol{\omega} \mid \mathcal{D})\, \mathrm{d}\boldsymbol{\omega}. \tag{3}$$

Para ResNet18, con $P \sim 10^7$ parámetros, ni el posterior $p(\boldsymbol{\omega} \mid \mathcal{D})$ ni la integral (3) admiten forma cerrada. La metodología de este trabajo se centra en cómo aproximar esa integral manteniendo el mayor número posible de garantías bayesianas a un coste computacional aceptable.

2.2 Requisitos del sistema

Requisitos funcionales. El sistema debe (i) emitir, para cada entrada, una distribución predictiva $p(y \mid \mathbf{x})$ en lugar de una estimación puntual; (ii) descomponer la incertidumbre en sus componentes aleatórica (inherente al ruido de los datos, irreducible) y epistémica (asociada al conocimiento limitado del modelo, reducible con más datos); y (iii) producir derivar una muestra a revisión humana cuando la incertidumbre epistémica de su predicción supere el umbral preconfigurado τ_u.

Requisitos no funcionales. Establecemos como objetivos cuantitativos (i) un *Expected Calibration Error* (ECE) por debajo de 0.05; (ii) una latencia de inferencia por imagen que sea compatible con la posible integración en una interfaz de un programa médico; y (iii) un consumo de memoria que permita la ejecución en hardware de una clínica u hospital estándar. Estos requisitos condicionan, como veremos en la SECCION2! la elección del subconjunto de pesos sobre el que aplicar Laplace y la estructura adoptada para el Hessiano.

2.3 Selección de métodos

La literatura ofrece un rango amplio de métodos de inferencia aproximada para redes neuronales bayesianas (BNNs). La Tabla 1 resume los factores que más comprometen la selección, la cual debe ponderar el coste de entrenamiento, el coste de inferencia, la necesidad o no de reentrenamiento y la fidelidad de la aproximación al posterior.

Table 1: Compromisos entre métodos de inferencia aproximada en BNN. T denota el número de muestras en inferencia y M el tamaño de un *ensemble.*

Método	Coste entren.	Coste inf.	Reentren.	Cobertura del posterior
Laplace (*post-hoc*)	$1\times$ + 1 *fit*	T	No	Local, unimodal
MC Dropout	$1\times$	T	Sí	Local, unimodal
Inferencia variacional (MFVI)	$\sim 2\times$	T	Sí	Unimodal
Deep Ensembles	$M\times$	M	Sí	Multimodal
MCMC (SGLD, HMC)	$\gg 1\times$	T	Sí	Asintóticamente exacta

SEGUIR DESDE AQUÍ

Seleccionamos **Laplace** y **MC Dropout** como representantes complementarios del espectro. El primero es *post-hoc* y *principled* —aprovecha la curvatura real del *loss* en el MAP— mientras que el segundo es trivial de implementar sobre cualquier arquitectura que ya incluya dropout. Ambos comparten la misma arquitectura troncal y, en el caso de Laplace, el mismo punto MAP, lo que permite atribuir las diferencias observadas casi exclusivamente al mecanismo de cuantificación de incertidumbre. Deep Ensembles y MCMC quedan fuera del alcance por coste computacional; MFVI se descarta por exigir una reimplementación completa del entrenamiento sin ventaja teórica clara sobre Laplace en términos de fidelidad local al posterior.

3 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points. Times New Roman is the preferred typeface throughout, and will be selected for you by default. Paragraphs are separated by ½ line space (5.5 points), with no indentation.

The paper title should be 17 point, initial caps/lower case, bold, centered between two horizontal rules. The top rule should be 4 points thick and the bottom rule should be 1 point thick. Allow ¼ inch space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the page.

For the final version, authors' names are set in boldface, and each name is centered above the corresponding address. The lead author's name is to be listed first (left-most), and the co-authors' names (if different address) are set to follow. If there is only one co-author, list both author and co-author side by side.

Please pay special attention to the instructions in Section **??** regarding figures, tables, acknowledgments, and references.

4 Metodología

4.1 Arquitectura troncal

Los tres modelos se construyen sobre una misma arquitectura base, concretamente una ResNet18 preentrenada en ImageNet, que actúa como extractor de características para la clasificación binaria de parches histopatológicos. Sea $g_{\boldsymbol{\phi}}(\mathbf{x}) \in \mathbb{R}^{512}$ la representación producida por el backbone para una imagen $\mathbf{x} \in \mathbb{R}^{3\times 96\times 96}$. Siguiendo una estrategia de *fine-tuning* parcial, congelamos los bloques `conv1`, `bn1`, `layer1` y `layer2`, y entrenamos únicamente `layer3`, `layer4` y la cabeza clasificadora. Esta decisión reduce el número de parámetros efectivamente optimizados y preserva representaciones visuales genéricas aprendidas en ImageNet.

El modelo determinista sustituye la capa totalmente conectada original por una proyección lineal

$$f_{\boldsymbol{\omega}}(\mathbf{x}) = \mathbf{W}\, g_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbf{b}, \qquad \mathbf{W} \in \mathbb{R}^{2\times 512}, \quad \mathbf{b} \in \mathbb{R}^{2}, \tag{4}$$

de forma que la salida son dos *logits* compatibles con la verosimilitud categórica de la Sección 2.1.

En MC Dropout mantenemos el mismo backbone, pero reemplazamos la cabeza por un pequeño bloque estocástico

$$h \mapsto \mathrm{Linear}_{256\to 2}\Big(\mathrm{Dropout}_p\big(\mathrm{ReLU}(\mathrm{Linear}_{512\to 256}(h))\big)\Big), \qquad p = 0.3, \tag{5}$$

introduciendo así la fuente de aleatoriedad necesaria para la aproximación variacional implícita.

4.2 Modelo determinista: estimación MAP

El clasificador de referencia se entrena minimizando la pérdida de entropía cruzada con regularización L_2:

$$\mathcal{L}(\boldsymbol{\omega}) = -\sum_{i=1}^{N} \log p(y_i \mid \mathbf{x}_i, \boldsymbol{\omega}) + \frac{\lambda}{2}\|\boldsymbol{\omega}\|_2^2, \qquad \lambda = 10^{-4}. \tag{6}$$

Bajo el prior gaussiano isótropo $p(\boldsymbol{\omega}) = \mathcal{N}(\mathbf{0}, \lambda^{-1}\mathbf{I})$, la minimización de (6) es equivalente a obtener el estimador MAP

$$\boldsymbol{\omega}^* = \arg\max_{\boldsymbol{\omega}} \left[\log p(\mathcal{D} \mid \boldsymbol{\omega}) + \log p(\boldsymbol{\omega})\right]. \tag{7}$$

Optimizamos (6) con Adam, tasa de aprendizaje inicial 10^{-4} y *weight decay* 10^{-4}. La selección del mejor checkpoint se realiza sobre el AUC de validación. Para estabilizar el entrenamiento usamos dos mecanismos: (i) *early stopping* con paciencia 5 sobre AUC de validación y (ii) `ReduceLROnPlateau` con factor 0.5 y paciencia 3. Cuando hay GPU disponible, el entrenamiento utiliza precisión mixta automática para acelerar el cálculo sin alterar apreciablemente las métricas finales.

Este modelo no es bayesiano en sentido pleno, pero es fundamental por dos motivos: primero, actúa como *baseline* fuerte; segundo, su punto MAP $\boldsymbol{\omega}^*$ es precisamente el centro alrededor del cual se construye la aproximación de Laplace.

4.3 Aproximación de Laplace

Aproximación local del posterior. La aproximación de Laplace reemplaza el posterior exacto $p(\boldsymbol{\omega} \mid \mathcal{D})$ por una Gaussiana centrada en el modo MAP. Si expandimos el log-posterior alrededor de $\boldsymbol{\omega}^*$ mediante Taylor de segundo orden, obtenemos

$$\log p(\boldsymbol{\omega} \mid \mathcal{D}) \approx \log p(\boldsymbol{\omega}^* \mid \mathcal{D}) - \frac{1}{2}(\boldsymbol{\omega} - \boldsymbol{\omega}^*)^\top \mathbf{H}(\boldsymbol{\omega} - \boldsymbol{\omega}^*), \tag{8}$$

donde $\mathbf{H} = -\nabla^2_{\boldsymbol{\omega}} \log p(\boldsymbol{\omega}^* \mid \mathcal{D})$. Exponenciando y renormalizando, el posterior aproximado es

$$q(\boldsymbol{\omega}) = \mathcal{N}\big(\boldsymbol{\omega} \mid \boldsymbol{\omega}^*, \boldsymbol{\Sigma}\big), \qquad \boldsymbol{\Sigma} = (\mathbf{H} + \lambda_{\mathrm{LA}}\mathbf{I})^{-1}, \tag{9}$$

donde λ_{LA} es la precisión del prior efectiva usada por Laplace. Nótese que distinguimos λ_{LA} del *weight decay* de entrenamiento: el primero se re-estima *post-hoc* maximizando verosimilitud marginal aproximada.

Laplace solo sobre la última capa. Aplicar (9) a todos los pesos de una ResNet18 completa es inviable. Por ello descomponemos $\boldsymbol{\omega} = (\boldsymbol{\phi}, \boldsymbol{\psi})$, donde $\boldsymbol{\phi}$ agrupa el backbone y $\boldsymbol{\psi} \in \mathbb{R}^{1026}$ los parámetros de la última capa lineal (512×2 pesos más 2 sesgos). El backbone se trata como extractor determinista fijo y la aproximación se restringe a

$$q(\boldsymbol{\psi}) = \mathcal{N}\big(\boldsymbol{\psi} \mid \boldsymbol{\psi}^*, \boldsymbol{\Sigma}_\psi\big). \tag{10}$$

Esta decisión conserva el grueso de la incertidumbre *task-specific* del clasificador con un coste muy inferior al de una Laplace de red completa.

Curvatura y estructura de Hessiano. En clasificación, el Hessiano exacto del log-posterior puede ser indefinido en redes profundas. Para obtener una matriz positiva semidefinida y numéricamente estable, empleamos la aproximación *Generalized Gauss–Newton* (GGN),

$$\mathbf{H}_{\mathrm{GGN}} = \sum_{i=1}^{N} \mathbf{J}_\psi(\mathbf{x}_i)^\top \boldsymbol{\Lambda}_i \mathbf{J}_\psi(\mathbf{x}_i), \tag{11}$$

donde $\mathbf{J}_\psi(\mathbf{x}_i)$ es el Jacobiano de los logits respecto a la última capa y $\mathbf{\Lambda}_i = -\nabla^2_{\mathbf{f}} \log p(y_i \mid \mathbf{f})$ es el Hessiano negativo de la log-verosimilitud respecto a los logits. Aun en la última capa, almacenar una covarianza densa puede ser innecesariamente costoso; por ello utilizamos la estructura Kronecker-factorizada (`kron`), que aproxima el Hessiano como producto de factores de menor dimensión y ofrece un compromiso favorable entre coste y fidelidad.

Ajuste de la precisión del prior. Tras ajustar Laplace sobre el conjunto de entrenamiento, optimizamos λ_{LA} maximizando una aproximación de la log-verosimilitud marginal,

$$\log p(\mathcal{D} \mid \lambda_{\mathrm{LA}}) \approx \log p(\mathcal{D} \mid \boldsymbol{\omega}^*) + \log p(\boldsymbol{\omega}^* \mid \lambda_{\mathrm{LA}}) - \frac{1}{2} \log \det\big(\mathbf{H}_{\mathrm{GGN}} + \lambda_{\mathrm{LA}} \mathbf{I}\big) + \mathrm{const}. \quad (12)$$

Esta etapa constituye una forma de selección de modelo bayesiana *post-hoc* y evita fijar manualmente la escala del prior de Laplace.

Predicción con Laplace. La distribución predictiva exacta bajo (10) es todavía intractable, por lo que recurrimos a dos aproximaciones complementarias. Para la probabilidad predictiva media final usamos la regla predictiva tipo GLM con aproximación *probit* sobre la salida linealizada de la última capa. Denotando por $\bar{p}_{\mathrm{LA}}(y = 1 \mid \mathbf{x}, \mathcal{D})$ dicha media,

$$\bar{p}_{\mathrm{LA}}(y = 1 \mid \mathbf{x}, \mathcal{D}) \approx p_{\text{GLM-probit}}(y = 1 \mid \mathbf{x}, \mathcal{D}). \quad (13)$$

Adicionalmente, para cuantificar incertidumbre epistémica extraemos $T_{\mathrm{LA}} = 100$ muestras posteriores $\boldsymbol{\psi}^{(t)} \sim q(\boldsymbol{\psi})$ y computamos

$$p_t(\mathbf{x}) = p\Big(y = 1 \mid \mathbf{x}, \boldsymbol{\psi}^{(t)}\Big), \qquad t = 1, \ldots, T_{\mathrm{LA}}. \quad (14)$$

Esta formulación híbrida separa explícitamente la regla predictiva empleada para calibración de la utilizada para estimar dispersión posterior.

4.4 MC Dropout como inferencia variacional

MC Dropout aproxima la integral predictiva bayesiana manteniendo activas las máscaras de dropout en inferencia. Siguiendo la interpretación variacional de **?**, cada máscara induce una realización de un posterior aproximado $q_{\mathrm{drop}}(\boldsymbol{\omega})$. En nuestro caso, la estocasticidad se introduce solo en la cabeza clasificadora de (5); el resto del backbone permanece determinista.

En tiempo de test fijamos la red en modo evaluación para no actualizar las estadísticas de BatchNorm, pero reactivamos exclusivamente las capas `Dropout`. De este modo, cada pasada hacia delante usa una máscara distinta y genera una muestra implícita del posterior aproximado. La predictiva se estima por Monte Carlo como

$$\bar{p}_{\mathrm{MC}}(y = 1 \mid \mathbf{x}, \mathcal{D}) \approx \frac{1}{T_{\mathrm{MC}}} \sum_{t=1}^{T_{\mathrm{MC}}} p\Big(y = 1 \mid \mathbf{x}, \boldsymbol{\omega}^{(t)}\Big), \qquad \boldsymbol{\omega}^{(t)} \sim q_{\mathrm{drop}}(\boldsymbol{\omega}), \quad (15)$$

con $T_{\mathrm{MC}} = 50$ pasadas estocásticas.

Frente a Laplace, MC Dropout presenta tres diferencias operativas relevantes. Primero, la familia variacional queda determinada por una probabilidad de dropout fija $p = 0.3$, que no se infiere del posterior. Segundo, el método requiere introducir dropout en arquitectura y entrenamiento, por lo que no es estrictamente *post-hoc*. Tercero, la incertidumbre se obtiene por muestreo funcional directo sin necesidad de construir explícitamente un Hessiano o una covarianza de pesos.

4.5 Descomposición de la incertidumbre y política de triaje

Para cada entrada $\mathbf{x}$ y cada muestra estocástica t definimos la probabilidad de clase positiva $p_t(\mathbf{x}) = p(y = 1 \mid \mathbf{x}, \boldsymbol{\omega}^{(t)})$. En MC Dropout empleamos la descomposición habitual de la varianza predictiva,

$$u_{\mathrm{epi}}(\mathbf{x}) = \mathrm{Var}_t[p_t(\mathbf{x})]\,, \quad (16)$$

$$u_{\mathrm{ale}}(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^{T} p_t(\mathbf{x})\big(1 - p_t(\mathbf{x})\big), \quad (17)$$

$$u_{\mathrm{tot}}(\mathbf{x}) = u_{\mathrm{epi}}(\mathbf{x}) + u_{\mathrm{ale}}(\mathbf{x}). \quad (18)$$

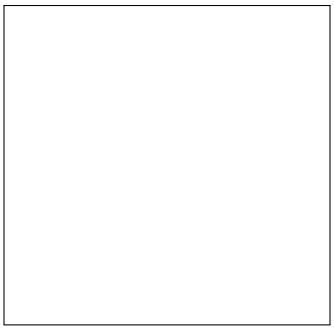

Figure 1: Sample figure caption.

La componente epistémica mide el desacuerdo entre muestras del posterior aproximado; la aleatórica captura la incertidumbre intrínseca de una Bernoulli condicional.

En Laplace usamos una variante coherente con la evaluación final del sistema. La incertidumbre epistémica se estima igualmente mediante $\mathrm{Var}_t[p_t(\mathbf{x})]$, pero la componente aleatórica se calcula a partir de la probabilidad media final obtenida por la regla GLM-probit:

$$u_{\mathrm{ale}}^{\mathrm{LA}}(\mathbf{x}) = \bar{p}_{\mathrm{LA}}(\mathbf{x})\,(1 - \bar{p}_{\mathrm{LA}}(\mathbf{x}))\,. \tag{19}$$

Con ello, la cantidad mostrada al usuario sigue siendo $u_{\mathrm{tot}} = u_{\mathrm{epi}} + u_{\mathrm{ale}}$, pero la media predictiva y la estimación de dispersión pueden provenir de aproximaciones complementarias.

Finalmente, traducimos la incertidumbre a una acción clínica mediante una política de rechazo selectivo. Una muestra se deriva a revisión humana cuando su incertidumbre epistémica excede un umbral τ_u calibrado en validación:

$$\mathrm{refer}(\mathbf{x}) = \mathbb{I}[u_{\mathrm{epi}}(\mathbf{x}) > \tau_u]\,. \tag{20}$$

En lugar de fijar un valor absoluto común a todos los métodos, estimamos τ_u como el percentil 85 de la distribución de incertidumbre epistémica en validación para cada aproximación. Esto evita comparar en una misma escala magnitudes cuya calibración numérica depende del mecanismo de inferencia empleado.

4.6 Footnotes

Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number[1] in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).

Note that footnotes are properly typeset *after* punctuation marks.[2]

4.7 Figures

All artwork must be neat, clean, and legible. Lines should be dark enough for purposes of reproduction. The figure number and caption always appear after the figure. Place one line space before the figure caption and one line space after the figure. The figure caption should be lower case (except for first word and proper nouns); figures are numbered consecutively.

You may use color figures. However, it is best for the figure captions and the paper body to be legible if the paper is printed in either black/white or in color.

4.8 Tables

All tables must be centered, neat, clean and legible. The table number and title always appear before the table. See Table 2.

[1] Sample of the first footnote.

[2] As in this example.

Table 2: Sample table title

	Part	
Name	Description	Size (μm)
Dendrite	Input terminal	$\sim$100
Axon	Output terminal	$\sim$10
Soma	Cell body	up to 10^6

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules.* We strongly suggest the use of the `booktabs` package, which allows for typesetting high-quality, professional tables:

`https://www.ctan.org/pkg/booktabs`

This package was used to typeset Table 2.

4.9 Math

Note that display math in bare TeX commands will not create correct line numbers for submission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You really shouldn't be using $$ anyway; see `https://tex.stackexchange.com/questions/503/why-is-preferable-to` and `https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath` for more information.)

4.10 Final instructions

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the **References** section; see below). Please note that pages should be numbered.

5 Preparing PDF files

Please prepare submission files with paper size "US Letter," and not, for example, "A4."

Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or Embedded TrueType fonts. Here are a few instructions to achieve this.

- You should directly generate PDF files using `pdflatex`.
- You can check which fonts a PDF files uses. In Acrobat Reader, select the menu Files>Document Properties>Fonts and select Show All Fonts. You can also use the program `pdffonts` which comes with `xpdf` and is available out-of-the-box on most Linux machines.
- `xfig` "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

  ```
  \usepackage{amsfonts}
  ```

  followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

  ```
  \newcommand{\RR}{I\!\!R} %real numbers
  \newcommand{\Nat}{I\!\!N} %natural numbers
  \newcommand{\CC}{I\!\!\!\!C} %complex numbers
  ```

  Note that `amsfonts` is automatically loaded by the `amssymb` package.

If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

5.1 Margins in LaTeX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below:

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (`http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf`)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command when necessary.

Acknowledgments and Disclosure of Funding

Use unnumbered first level headings for the acknowledgments. All acknowledgments go at the end of the paper before the list of references. Moreover, you are required to declare funding (financial activities supporting the submitted work) and competing interests (related financial activities outside the submitted work). More information about this disclosure can be found at: `https://neurips.cc/Conferences/2025/PaperInformation/FundingDisclosure`.

Do **not** include this section in the anonymized submission, only in the final paper. You can use the `ack` environment provided in the style file to automatically hide this section in the anonymized submission.

References

References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to `small` (9 point) when listing the references. Note that the Reference section does not count towards the page limit.

[1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.

[2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural SImulation System.* New York: TELOS/Springer–Verlag.

[3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

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